Why Domain Matters, part 2

Logistics and supply chain

The record says

`Delivered on time, in full.`

What actually happened

One pallet line held the wrong pack size.

Monday 22 June, 6:12 am. Store 014, Jayanagar.

The truck made its slot. Every record says the delivery was in full.

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Goods receipt GRN-260622, 06:12
Pallet …0542 scanned, notice matched
Line 1: atta 5 kg, 16 cases, 64 bags
Discrepancies: none
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On line 1, the 16 bales hold 1 kg packs, 160 of them. The 5 kg bags are still in the warehouse.

Five readings of one delivery

Who	What they see	What they conclude
DC shift supervisor	Every line picked and confirmed by a scan; two count variances in the atta bay, both corrected	“A clean night’s picking.”
Transport manager	Left 04:30, arrived 06:12, inside the slot; 23 of 25 mornings on time	“TRK-07 made its slot.”
Receiving clerk	Pallet labels matched the shipping notice	“No discrepancies.”
OTIF scorecard	88% for June; misses on 11, 13, 15 June, none of them atta	“22 June was perfect.”
Stock controller	26 June count: 5 kg −63, 1 kg +157	“Sloppy counting, or theft.”

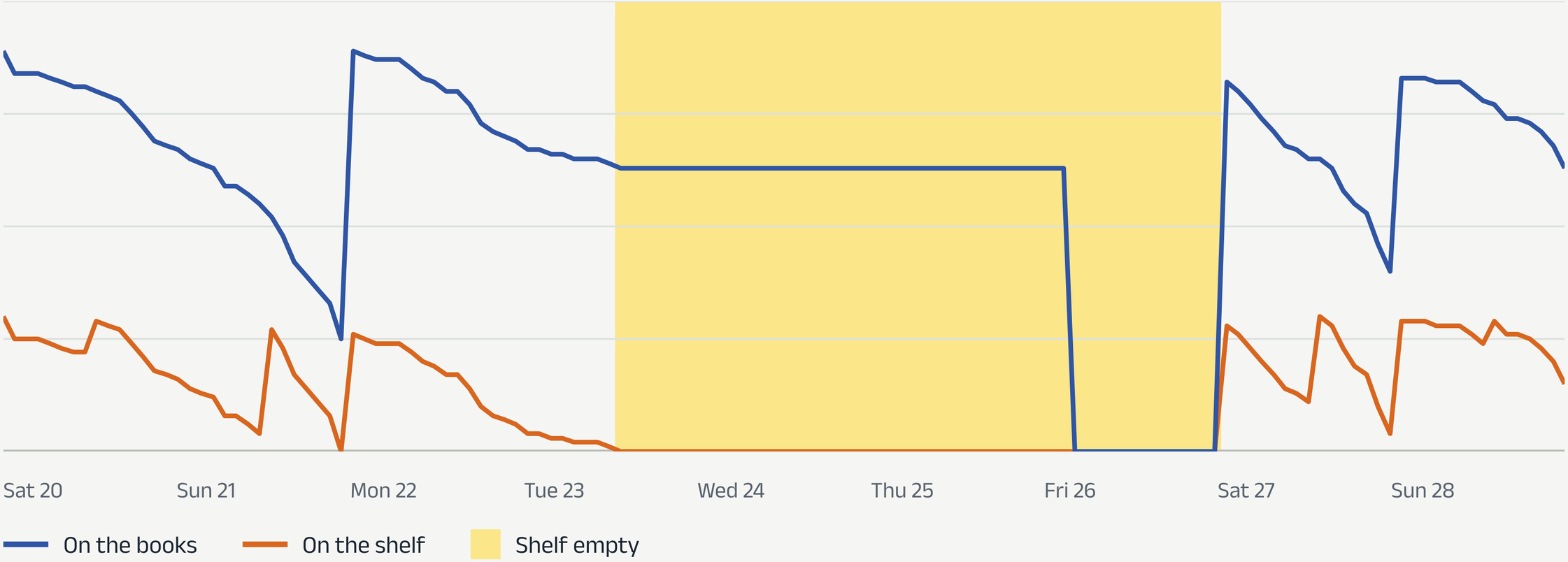
Each conclusion is reasonable given the record in front of that person. Each misses the wrong line.

The scan confirmed the slot, not the item

Pick confirmation	`03:00, slot A-07-03: 16 cases of atta 5 kg, by slot label scan`
Shipping notice	`Line 1: 16 cases of atta 5 kg, 64 bags`
Pallet label	`SSCC 029000010000010542`
Goods receipt	`GRN-260622, 06:12: 64 bags of atta 5 kg booked`
On the pallet	16 bales of atta 1 kg, 160 packs

At 03:00 the picker stood at slot A-07-03 (5 kg) and took the bales from A-07-04 next to it. Every record after that was written from the pick confirmation, so they all agree.

The books said 63. The shelf was empty.



5 kg atta at Store 014, bags, every open hour from 20 to 28 June 2026. Grid lines every 25 bags. Synthetic world, seed 42.

What the shelf lost

51.5

open hours with the 5 kg shelf empty

₹19,080

of 5 kg atta customers wanted and could not buy

72 bags wanted during the gap and not bought

What the shopper did	Bags
Went to another store	29
Took a different pack size	16
Bought another brand	12
Came back later	9
Never bought it	6

At the shelf price of ₹265 a bag. Shopper responses follow Gruen, Corsten and Bharadwaj (2002).

Four corrections, one mistake

Where and when	Item	Units	At cost	Recorded as
Warehouse, Tue 23 Jun 14:20	Atta 1 kg	−160 (−16 cases)	−₹7,920	slot count variance
Warehouse, Thu 25 Jun 10:00	Atta 5 kg	+64 (+16 cases)	₹14,592	cycle count variance
Store, Fri 26 Jun 09:00	Atta 5 kg	−63	−₹14,364	count after complaints
Store, Fri 26 Jun 09:00	Atta 1 kg	+157	₹7,772	count after complaints
Net, across the company			₹80	

Equal and opposite variances in neighbouring slots, mirrored at a store that got a truck from them, is the signature of a mis-pick.

This is the normal case

99.68%

best-in-class order-picking accuracy (WERC DC Measures, 2025). About one in 300 still goes wrong.

57%

of supplier orders physically checked at retail warehouses, on average; about 1 retailer in 10 checks every unit

92%

median accuracy those checks found, against a median goal of 98%

Receiving by scanning a pallet's SSCC against the shipping notice is how GS1's standards are meant to work. It is exactly as accurate as the notice.

Sources: WERC DC Measures 2025 via Yale; Gibson (2017), Auburn University ASN study; GS1 UK, Logistics labels and SSCCs.

One name, four numbers

92%

The DC's report

Dispatched on schedule, shipped as ordered, by its own pick confirmations

90%

Scorecard, per line

Order lines in the slot and in full, by the store's goods receipt

88%

Scorecard, per delivery

Deliveries in the slot with every line in full, by goods receipt

84%

What the shelf got

Deliveries in the slot with every case what was ordered

OTIF for the warehouse's 25 deliveries to Store 014 in June. The 22 June delivery passes the first three and fails the fourth, which nobody computes because nobody records it.

Two of three fixes make OTIF worse

Scan every case at pick

88% → 88%

scorecard unchanged, and now true: the shelf got 88% too; 1 bag missed all week

The scanner reads the bale, not the slot, and rejects 1 kg on a 5 kg line.

Weigh every pallet

88% → 84%

scorecard falls; 1 bag missed all week

The pallet is 160 kg light. Re-picked, the truck arrives 06:42, late.

Scan every case at the store

88% → 84%

scorecard falls; 0 bags missed

The clerk books what arrived and raises a claim. 5 kg is reordered that night.

OTIF per delivery for June, on the scorecard. Same customers and trucks in every scenario; the changes start on 1 June. Try them in the demo.

Nothing in the records says “wrong item”

Pallet	Line	Item	Cases	Units
…0542	1	Atta 5 kg	16	64
…0542	2	Atta 1 kg	9	90
…0542	3	Rice 5 kg	14	56
…0559	4	Sugar 1 kg	5	50

Receipt: booked as notice. Discrepancies: none.

The notice, the pallet label and the receipt join without a mismatch, because all three come from the same pick.

The evidence is in records that do not share that source: pallet weights against case weights, a 1 kg backroom growing without receipts, 5 kg sales stopping while the books sit at 63.

Shipping notice SHP-260622 and goods receipt GRN-260622. Identifiers use GS1 prefix 29, set aside for restricted circulation.

How to learn this domain

How a warehouse works	Bartholdi and Hackman, *Warehouse & Distribution Science* (free from the authors), and Gwynne Richards, *Warehouse Management*. Then walk a warehouse, dock to dispatch.
Fundamentals	MIT's Supply Chain Fundamentals (CTL.SC1x): demand, inventory and transport.
Documents and IDs	GS1 logistic label, SSCC and despatch advice. In India, the e-way bill for most consignments over ₹50,000 (rule 138, CGST Rules).
Metrics	OTIF, case fill, pick accuracy, dock-to-stock. How Walmart's OTIF changed; WERC's DC Measures.
Practice	Download June's shipping records, compute OTIF two ways, and find the 22 June line.

A rough plan for a few evenings a week over about two months. Links in the post: kvinayakpai.github.io/domain-series/logistics

Every record agreed. That is why nobody checked the pallet.

Run the scenarios and download the records:

`kvinayakpai.github.io/domain-series`

> **Next: payments**
>
> The till says the payment failed. The customer’s bank says it went through.